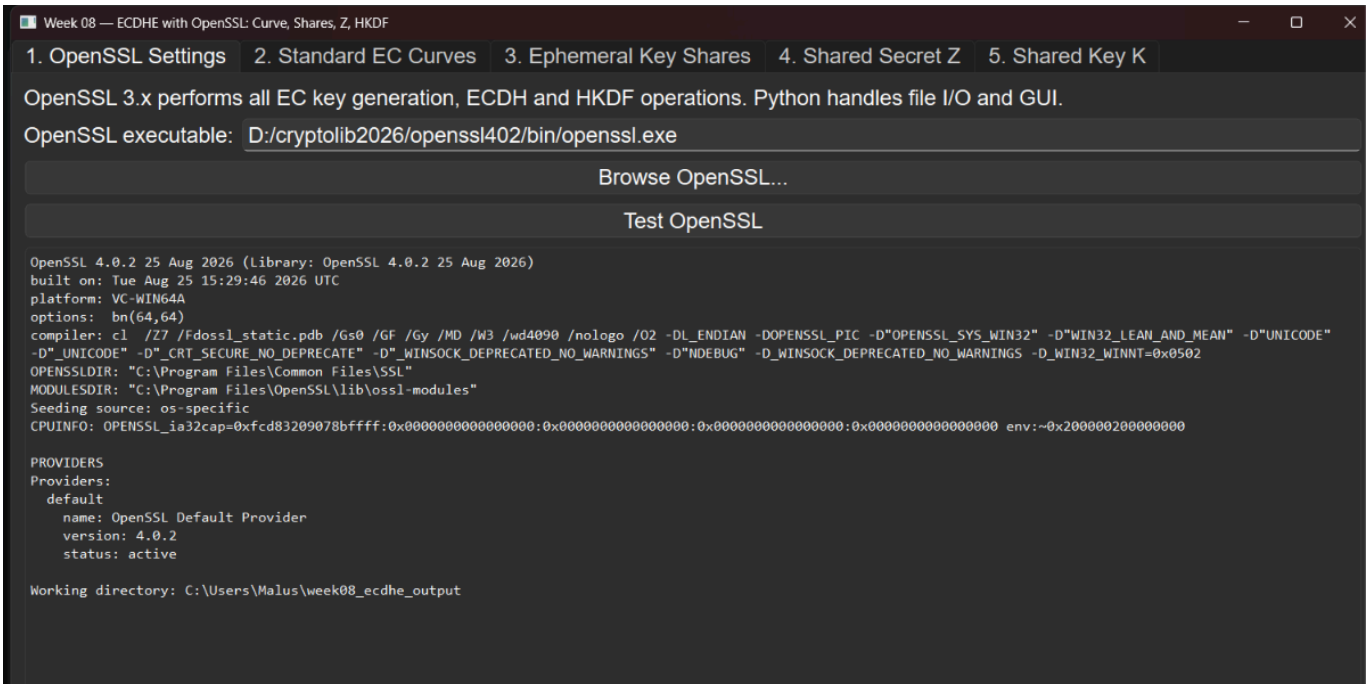
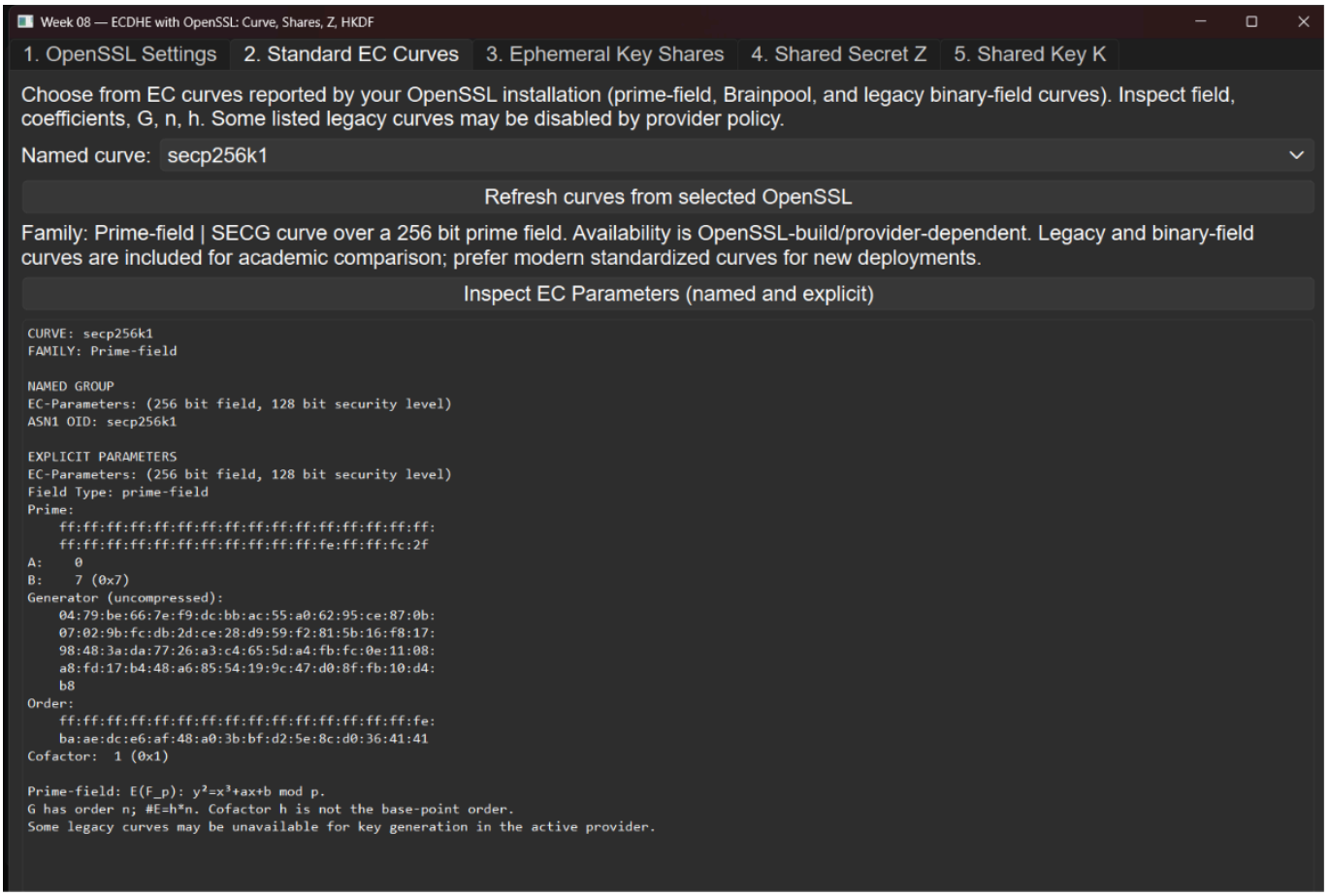
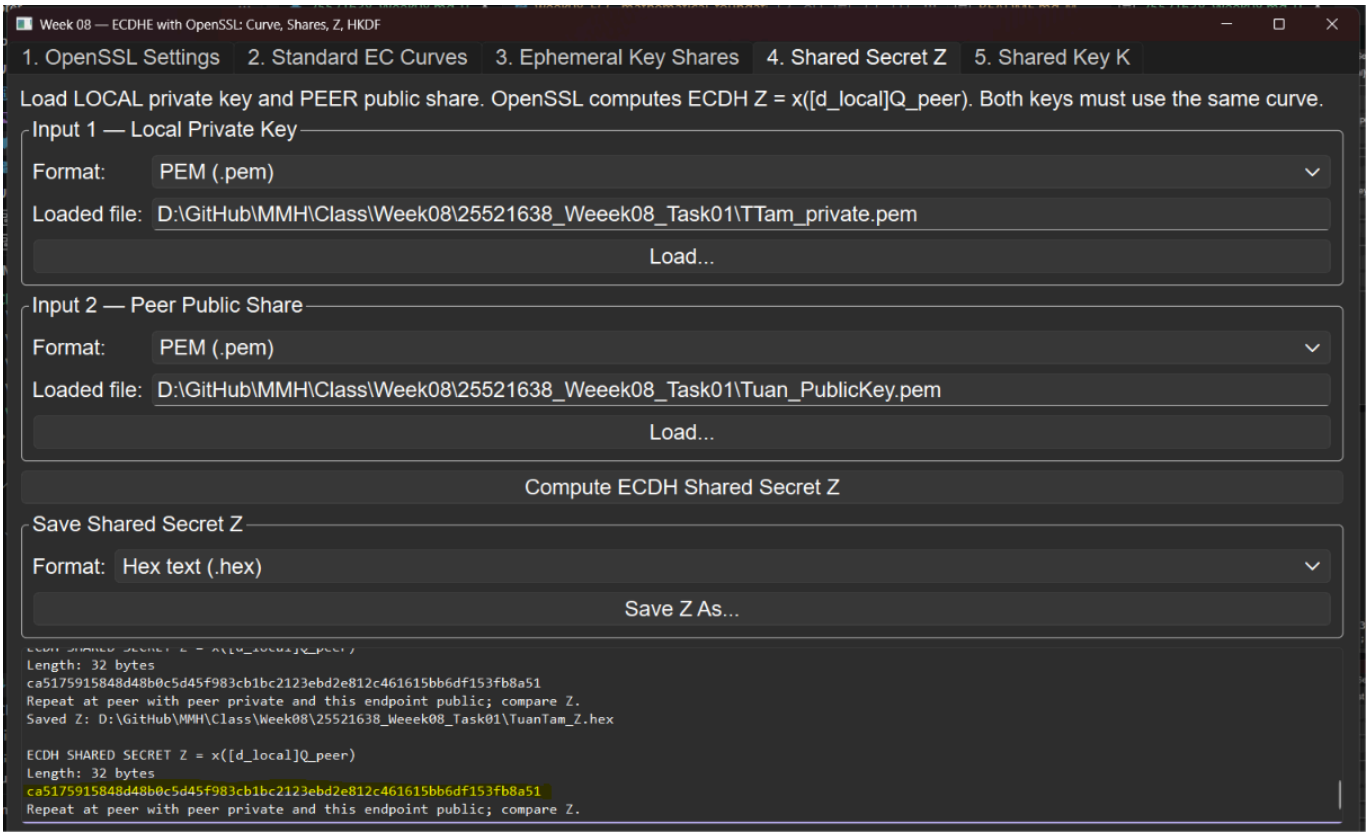
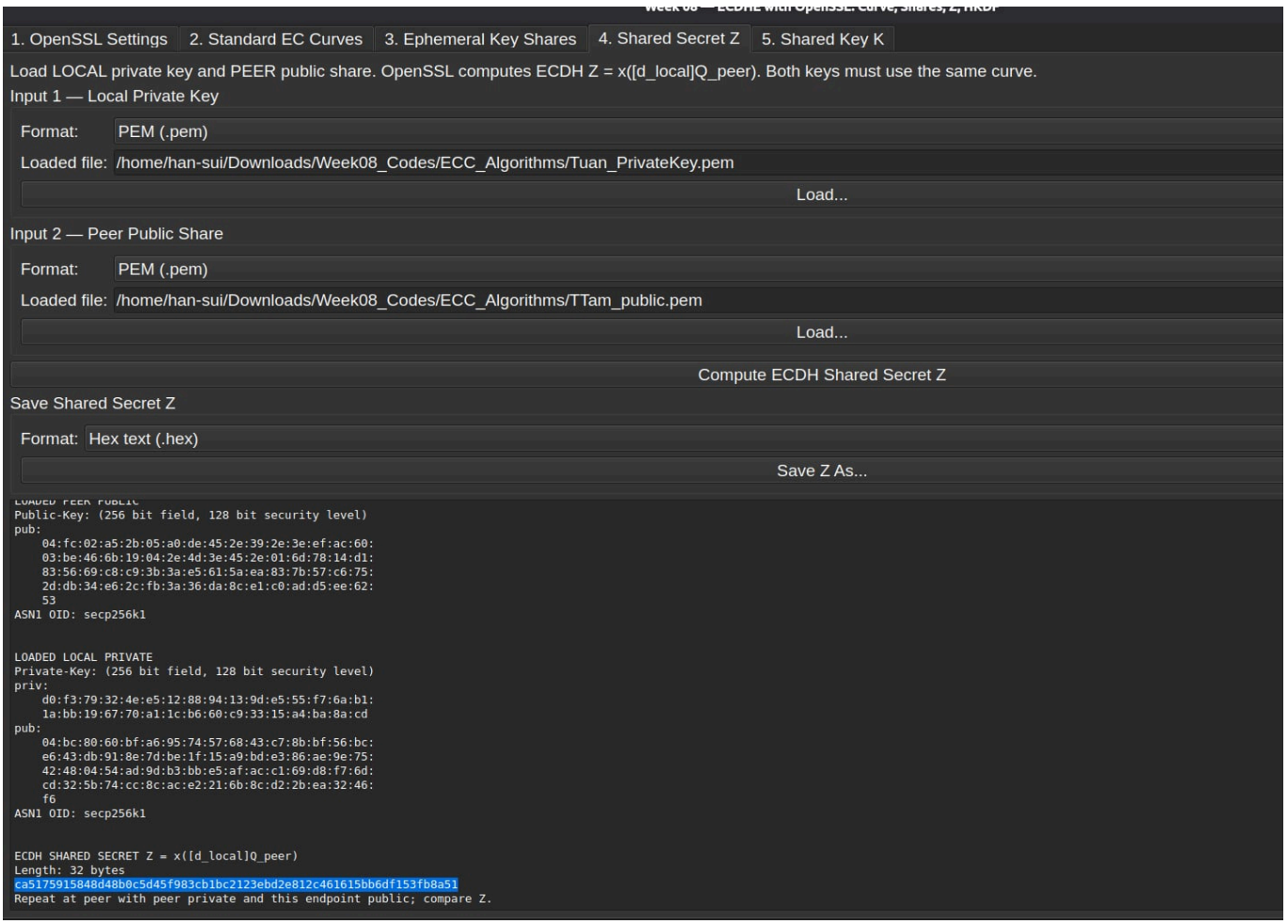
Ver openssl: 4.0.2

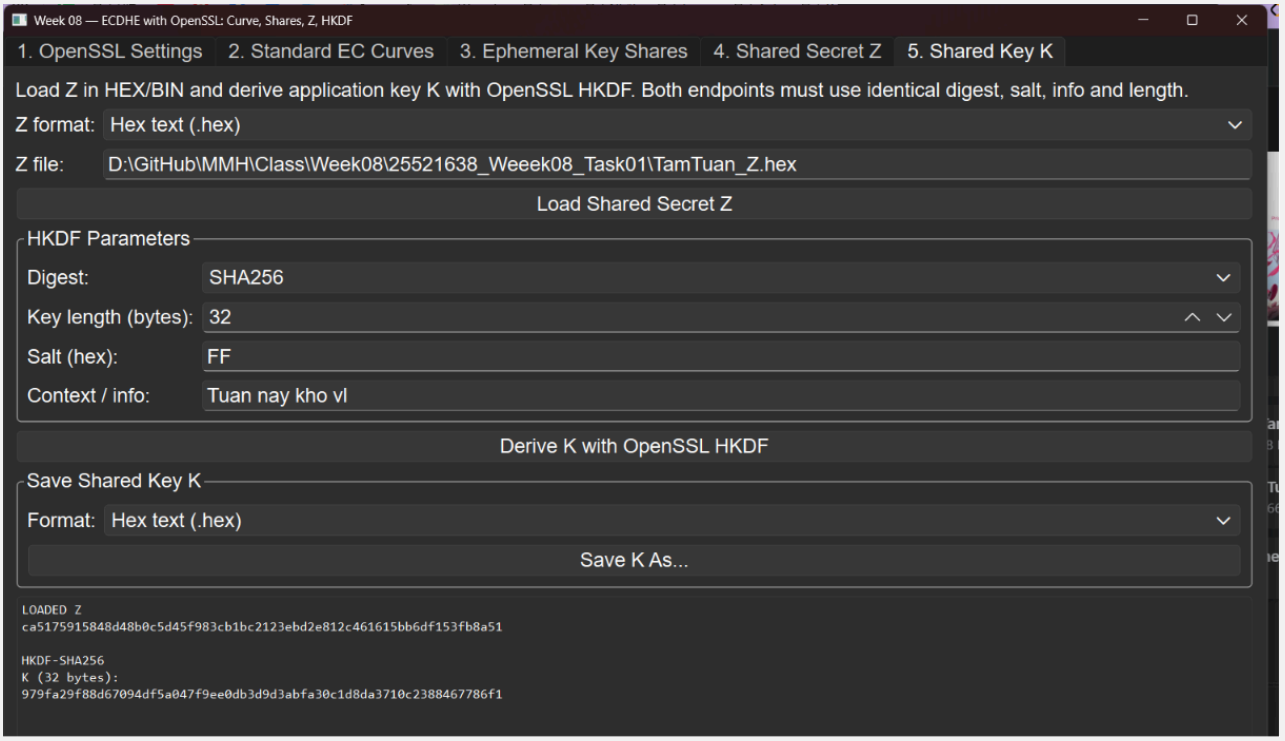


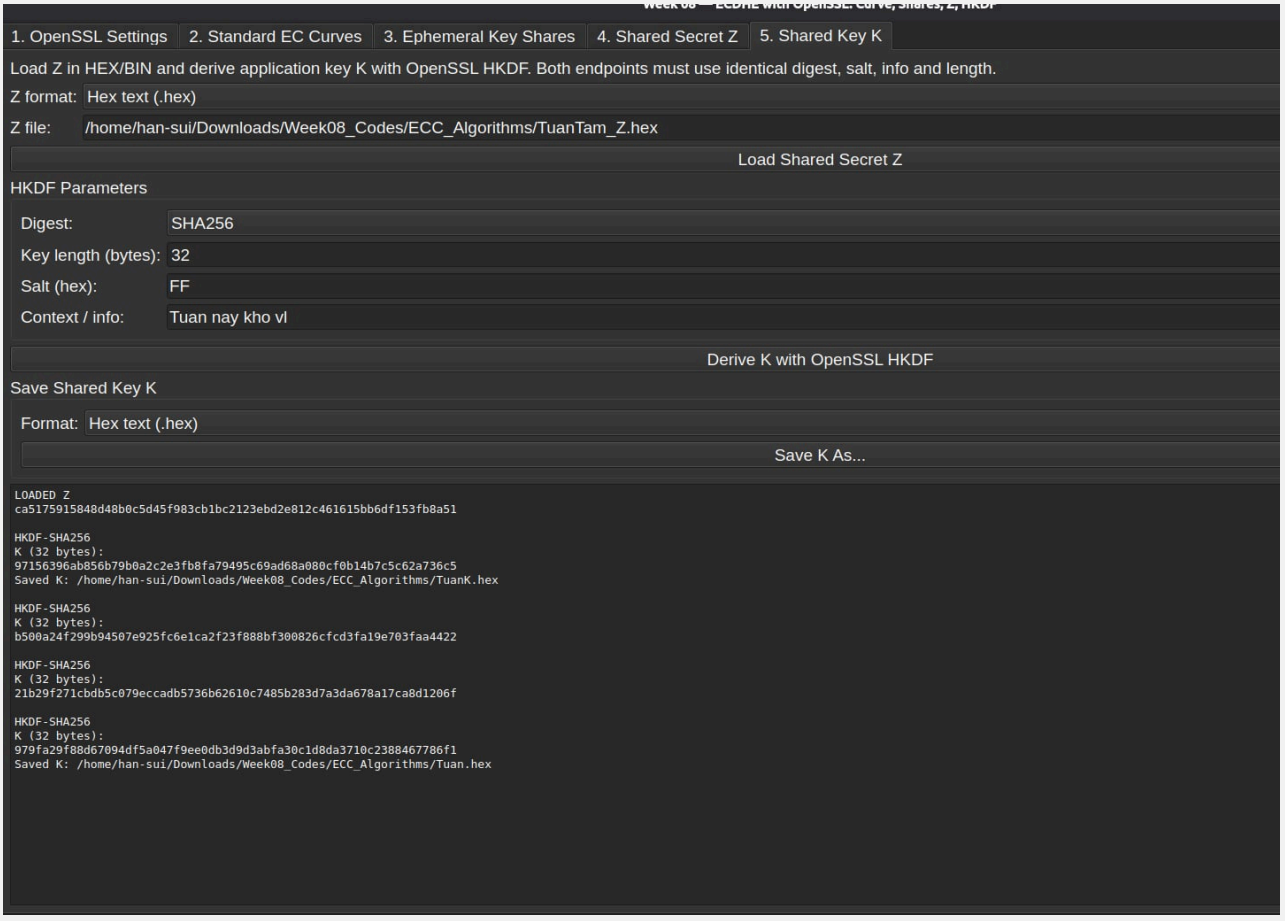



Week 08 — ECDHE with OpenSSL: Curve, Shares, Z, HKDF
1. OpenSSL Settings
2. Standard EC Curves
3. Ephemeral Key Shares
4. Shared Secret Z
5. Shared Key K
Load LOCAL private key and PEER public share. OpenSSL computes ECDH Z = x([d_local]Q_peer). Both keys must use the same curve.
Input 1 — Local Private Key
Format: PEM (.pem)
Loaded file: D:\GitHub\MMH\Class\Week08\25521638_Weeek08_Task01\TTam_private.pem
Load...
Input 2 — Peer Public Share
Format: PEM (.pem)
Loaded file: D:\GitHub\MMH\Class\Week08\25521638_Weeek08_Task01\Tuan_PublicKey.pem
Load...
Compute ECDH Shared Secret Z
Save Shared Secret Z
Format: Hex text (.hex)
Save Z As...
Length: 32 bytes
ca5175915848d48b0c5d45f983cb1bc2123ebd2e812c461615bb6df153fb8a51
Repeat at peer with peer private and this endpoint public; compare Z.
Saved Z: D:\GitHub\MMH\Class\Week08\25521638_Weeek08_Task01\TuanTam_Z.hex
ECDH SHARED SECRET Z = x([d_local]Q_peer)
Length: 32 bytes
ca5175915848d48b0c5d45f983cb1bc2123ebd2e812c461615bb6df153fb8a51
Repeat at peer with peer private and this endpoint public; compare Z.


1. OpenSSL Settings
2. Standard EC Curves
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5. Shared Key K
Load LOCAL private key and PEER public share. OpenSSL computes ECDH Z = x([d_local]Q_peer). Both keys must use the same curve.
Input 1 — Local Private Key
Format: PEM (.pem)
Loaded file: /home/han-sui/Downloads/Week08_Codes/ECC_Algorithms/Tuan_PrivateKey.pem
Load...
Input 2 — Peer Public Share
Format: PEM (.pem)
Loaded file: /home/han-sui/Downloads/Week08_Codes/ECC_Algorithms/TTam_public.pem
Load...
Compute ECDH Shared Secret Z
Save Shared Secret Z
Format: Hex text (.hex)
Save Z As...
Public-Key: (256 bit field, 128 bit security level)
pub:
04:fc:02:a5:2b:05:a0:de:45:2e:39:2e:3e:ef:ac:60:
03:be:46:6b:19:04:2e:4d:3e:45:2e:01:6d:78:14:d1:
83:56:69:c8:c9:3b:3a:e5:61:5a:ea:83:7b:57:c6:75:
2d:db:34:e6:2c:fb:3a:36:da:8c:e1:c0:ad:d5:ee:62:
53
ASN1 OID: secp256k1
LOADED LOCAL PRIVATE
Private-Key: (256 bit field, 128 bit security level)
priv:
d0:f3:79:32:4e:e5:12:88:94:13:9d:e5:55:f7:6a:b1:
1a:bb:19:67:70:a1:1c:b6:60:c9:33:15:a4:ba:8a:cd
pub:
04:bc:80:60:bf:a6:95:74:57:68:43:c7:8b:bf:56:bc:
e6:43:db:91:8e:7d:be:1f:15:a9:bd:e3:86:ae:9e:75:
42:48:04:54:ad:9d:b3:bb:e5:af:ac:c1:69:d8:f7:6d:
cd:32:5b:74:cc:8c:ac:e2:21:6b:8c:d2:2b:ea:32:46:
f6
ASN1 OID: secp256k1
ECDH SHARED SECRET Z = x([d_local]Q_peer)
Length: 32 bytes
ca5175915848d48b0c5d45f983cb1bc2123ebd2e812c461615bb6df153fb8a51
Repeat at peer with peer private and this endpoint public; compare Z.

Key Z giống nhau





Khi cho hai thông tin giống nhau sẽ ra output giống nhau